



Valentin Legras

Senior Data Scientist

Basel, Switzerland

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Profile

Senior Data Scientist with 7+ years in clinical pharmacometrics and PK/PD data science across Roche, Novartis, and Sanofi. Founded and led aNCA, an open-source R package adopted by 120+ scientists at Roche, delivering more than 2M CHF in annual savings. Pairs regulatory-grade R, NONMEM, and CDISC (SDTM/ADaM) data engineering with Agile product ownership and an optimization-first mindset. Scales AI and multi-agent coding workflows to automate complex analysis and accelerate delivery, supporting molecules from Phase 1 to market.

Work Experience

Senior Data Scientist

2020 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Engineered end-to-end PK/PD analysis workflows, from data acquisition and preparation to modeling and reporting, across multiple therapeutic areas (e.g., Oncology, Neuroscience) for Phase 1 to 3 clinical trials.
- Developed interactive R/Shiny and Spotfire tools for real-time data review, enabling faster cross-functional decisions on complex clinical data.
- Led the creation of analysis-ready NCA, PK/PD, and ADaM-compliant datasets from raw clinical data for population modeling in R, Phoenix and NONMEM.
- Authored responses to health authority queries (FDA, EMA) and managed the QC and preparation of datasets and reports for electronic regulatory submissions, contributing to the successful progression of molecules from Phase 1 to market.
- Led a process improvement initiative: automation of ADPC, ADNCA, and ADPP datasets by converting legacy SAS programs into reproducible R pipelines, reducing manual effort by 90%. This new process is now used by 30 data scientists, saving them 70% of their time on these tasks.
- Scaled multi-agent AI coding workflows across the team, including AI-augmented issue creation, automated problem solving, and code review, boosting development velocity while upholding high engineering standards.
- Partnered with clinical, statistical, and pharmacometric teams to ensure data integrity and deliver custom tools that streamlined modeling and simulation.

aNCA Product Owner

2023 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Directed the end-to-end product lifecycle of the 'aNCA' open-source R package under Agile/Scrum, driving it from concept to enterprise-wide adoption by 120+ clinical and pre-clinical scientists at Roche.
- Led, coached, and mentored a core team of 7 developers, onboarded new clinical and pre-clinical colleagues, and managed a 15+ stakeholder network to define a product roadmap now supporting 2,000+ studies annually.
- Delivered more than 2M CHF in annual savings by transforming a manual, weeks-long workflow into a 90% automated process. This impact combines 5 FTEs and retired commercial software licenses.
- Established and enforced GxP-compliant R package development standards, including rigorous code review and mentorship, to ensure regulatory-grade quality.
- Contributed to the [pharmaverse](#) ecosystem by developing standardized modules for automated NCA, PK Table, Listing, and Graph (TLG) generation, using Git, CI/CD, Coding Agents (LLMs) and an AI interpretation bot (Python).

IT/Data Science Developer Internship

Mar 2020 to Aug 2020

F. Hoffmann-La Roche, Basel, Switzerland

- Translated pharma scientists' requirements into IT/Data science tasks.
- Deployed several tools with R Shiny (data visualization) to support process modelers and formulators (incl. tidyverse packages).
- Used Git, Jenkins and GCP to build, test and deploy applications.
- Automated database updates on Google Spreadsheet with R (incl. VBA coding), data management and data merging.
- Automated PowerPoint reports with R (officer package) for daily reports.

Data Scientist Apprenticeship

Oct 2019 to Feb 2020

Cofidis, Lille, France

- Analyzed customer loan data in R and built machine learning models (Logistic Regression, Gradient Boosting, Random Forest) to predict contract default.
- Recommended data collection strategies to optimize the rate of returned and completed contracts.

Biostatistical Application Developer

Internship



May 2019 to Aug 2019

Novartis AG, Basel, Switzerland

- Built an R Shiny application to visualize clinical data (tables, figures and listings) interactively during DMC (Data Monitoring Committee) reviews, and generalized it across studies.
- Created dynamic HTML reports with R Markdown (dplyr, ggplot, plotly, DT) using ADaM and SDTM datasets (CDISC formats).

SAS Developer

Internship



Jun 2018 to Aug 2018

Sanofi R&D, Paris, France

- Developed and validated SAS (BASE/MACRO/SQL) programs for TLG reporting and built a data-mapping tool generating XML for an AI project, using SDTM/ADaM specifications.

SAS and IT Developer

Internship



Apr 2017 to Aug 2017

Sanofi R&D, Paris, France

- Migrated and optimized SAS programs for clinical trial data in a new statistical environment, enabling biostatisticians to run their analyses.

Skills

Pharmacometrics & Clinical Data R, NONMEM, Phoenix/WinNonlin, PK/PD modeling, NCA, CDISC (SDTM/ADaM), biostatistics, GxP	Visualization & Reporting R Shiny, ggplot, R Markdown, SAS, Spotfire/Tableau
Data Science & Engineering Python, machine learning, SQL, Git, CI/CD, Agile/Scrum	Leadership & Delivery team leadership, mentoring, product ownership, stakeholder management, cross-functional collaboration
AI & Automation Claude (Sonnet/Opus), coding agents (Ona, Claude Code), multi-agent workflows, LLM-assisted development, prompt engineering, AI-driven automation pipelines	Languages English (Fluent) ●●●●● French (Native) ●●●●● Spanish (B2) ●●●○○

Educational Background

Master's Degree, Software Engineering and Statistics

2017 to 2020

Polytech Lille, Villeneuve-d'Ascq, France

Completed alongside internships and an apprenticeship (work-study) at Sanofi, Novartis, Cofidis, and Roche.

Technical Degree, Statistics and Business Intelligence

2015 to 2017

Institut Universitaire de Technologie, Vannes, France

Publications and Presentations

Peer-Reviewed Publications

- [Computer-Aided formulation design for pharmaceutical drug product development, part 01: Materials exploration through a visualization tool](#). Piccione P.M., Lang M.N., Amado Becker F., Hofstetter A., Marchal S., Ly K., Legras V., Ewert A., Kohler D., Maurer R., Willecke N., Burwood R., Kroll P. (2024), International Journal of Pharmaceutics, Volume 667, Part B.
- [PBPK Modeling of Entrectinib and Its Active Metabolite to Derive Dose Adjustments in Pediatric Populations Co-Administered with CYP3A4 Inhibitors](#). Umehara K., Parrott N., Schindler E., Legras V. and Meneses-Lorente G. (2024), Clin Pharmacol Ther, 116: 1130-1140.
- [Dexamethasone is associated with reduced frequency and intensity of cytokine release syndrome compared with alternative corticosteroid regimens as premedication for glofitamab in patients with relapsed/refractory large B-cell lymphoma](#). Haematologica (2024). Contributed sunburst visualization analysis.
- Sunburst visualization method contributed to [Hipp J.F. et al., The UBE3A-ATS antisense oligonucleotide rugonersen in children with Angelman syndrome: a phase 1 trial](#). Nat Med 2025; 31(9):2936-2945.

Conference Presentations

- [Revolutionizing Data Exploration: An Interactive Journey through R Shiny](#). Legras V., PHUSE 2023.
- [R Scripts to Streamline Non-Compartmental Analysis \(NCA\)](#). Legras V., Suekuer E., PHUSE 2024 (aNCA open-source).
- [Sunburst Plots for Analysing the Impact of Drugs on Cytokine Release Syndrome \(CRS\) Management](#). Legras V., PAGE 2024.
- [Automated QC Framework for Reliable ADNCA Dataset Generation with admiral](#). Legras V., Aspridis L., PHUSE 2025.

Hobbies

Roller skating, Ice skating, Surfing, Basketball, Cycling, Drawing, Coding ([stock market analysis](#)).